

Clinical Risk Assessment Report: Patient 64

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.44x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.5%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -2.29 [Bottom 2%] (-2.5%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.16 [Avg (59th %ile)] (-1.0%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.62 [Bottom 21%] (-0.8%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.26 [Top 11%] (-0.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.86 [Bottom 16%] (-0.6%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.11 [Avg (58th %ile)] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.44 [Bottom 23%] (-0.6%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.58 [Avg (25th %ile)] (-0.6%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Machine Learning Feature Attribution (SHAP Decision Path)

